Girish

girish.research.pr@gmail.com | +91-9588703587 | Website | girish-b794092a1 | gir-ish
Google Scholar

Education

UPES **Bachelor of Technology in Computer Science and Engineering** (AI-ML Hons.) — Aug 2022 – May 2026

- CGPA: 7.99/10.0

Research Interests: *Multimodal Learning, Audio LLM, Speech Deepfake, Speech + Healthcare, Speech Processing.*
Languages: *English (IELTS: 6.5), Hindi (Native)*

Experiences

IIIT-Delhi, Under-graduate Research Assistant — New Delhi, India *(Hybrid)* June 2024 – Present
Advisor : Dr. Arun Balaji Buduru, Orchid Chetia Phukan

- Researched deepfake detection, speech emotion recognition, multimodal AI, and Audio Language Models (ALM); collaborated with domain experts to improve ML/DL models for real-world deployment.

Ulster University, Under-graduate Research Assistant — London Derry, UK *(Remote)* Dec 2023 – Present
Advisor : Dr. Muskaan Singh

- Conducted NLP and speech analysis research for autism detection using code-switched speech; improved algorithms, analyzed multilingual datasets, and enhanced model performance for early diagnosis.

Reliance Jio AICoE, ML Engineer — Hyderabad, India *(Remote)* May 2024 – June 2024

- Evaluated Speech Foundation Models (SFMs) for stress recognition, showing multilingual SFMs outperform raw-data models for physiological signal understanding; worked with interdisciplinary teams to validate cross-domain generalization of advanced speech models beyond speech tasks.

ARTVIEWINGS LLC, AI Research Engineer — San Jose, California *(Remote)* July 2024 – Aug 2024

- Developed a scalable AI system for multilingual audio-visual question answering; integrated the MERA framework with ensemble techniques and multilingual data to improve performance across eight languages.

Suratec Co., LTD, Computer Vision Intern — Bangkok, Thailand *(Remote)* June 2024 – July 2024

- Built an application to detect and analyze golf swing phases using video processing and ML; designed a real-time feedback interface and optimized handling of multiple video formats to improve accuracy and user experience.

Publications

Authors marked with an asterisk contributed equally to this work as first authors.

1. **Bridging Attribution and Open-Set Detection using Graph-Augmented Instance Learning in Synthetic Speech**
   Mohd Mujtaba Akhtar*, **Girish***, Farhan Sheth, Muskaan Singh
   EACL 2026 Main
2. **DIVINE: Coordinating Multimodal Disentangled Representations for Oro-Facial Neurological Disorder Assessment**
   Mohd Mujtaba Akhtar*, **Girish***, Muskaan Singh

EACL 2026 Main **Social Impact Award**

3. **Towards Attribution of Generators and Emotional Manipulation in Cross-Lingual Synthetic Speech using Geometric Learning**

   Girish*, Mohd Mujtaba Akhtar*, Farhan Sheth*, Muskaan Singh

   IJCNLP-AACL 2025 Findings

4. **Curved Worlds, Clear Boundaries: Generalizing Speech Deepfake Detection using Hyperbolic and Spherical Geometry Spaces**

   Farhan Sheth*, **Girish***, Mohd Mujtaba Akhtar*, Muskaan Singh

   IJCNLP-AACL 2025 Main

5. **HYFuse: Aligning Heterogeneous Speech Pre-Trained Representations in Hyperbolic Space for Speech Emotion Recognition**

   Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Swarup Ranjan Behera, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

   INTERSPEECH 2025 **Oral**

6. **Investigating the Reasonable Effectiveness of Speaker Pre-Trained Models and their Synergistic Power for SingMOS Prediction**

   Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Swarup Ranjan Behera, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

   INTERSPEECH 2025

7. **Towards Source Attribution of Singing Voice Deepfake with Multimodal Foundation Models**

   Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Swarup Ranjan, BeheraPriyabrata Mallick, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

   INTERSPEECH 2025

8. **PARROT: Synergizing Mamba and Attention-based SSL Pre-Trained Models via Parallel Branch Hadamard Optimal Transport for Speech Emotion Recognition**

   Orchid Chetia Phukan *, Mohd Mujtaba Akhtar *, **Girish** *, Swarup Ranjan Behera, Sai Kiran Patibandla, Arun Balaji Buduru, Rajesh Sharma

   INTERSPEECH 2025

9. **SNIFR: Boosting Fine-Grained Child Harmful Content Detection Through Audio-Visual Alignment with Cascaded Cross-Transformer**

   Orchid Chetia Phukan *, Mohd Mujtaba Akhtar *, **Girish** *, Swarup Ranjan Behera, Abu Osama Siddiqui, Sarthak Jain, Priyabrata Mallick, Sai Kiran Patibandla, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

   INTERSPEECH 2025

10. **Towards Machine Unlearning for Paralinguistic Speech Processing**

    Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Shubham Singh, Swarup Ranjan Behera, Vandana Rajan, Muskaan Singh, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025

11. **Towards Fusion of Neural Audio Codec-based Representations with Spectral for Heart Murmur Classification via Bandit-based Cross-Attention Mechanism**

    Orchid Chetia Phukan *, **Girish** *, Mohd Mujtaba Akhtar *, Swarup Ranjan Behera, Priyabrata Mallick, Santanu Roy, Arun Balaji Buduru, Rajesh Sharma

    INTERSPEECH 2025

12. **SOURCE TRACING OF SYNTHETIC SPEECH SYSTEMS THROUGH PARALINGUISTIC PRE-TRAINED REPRESENTATIONS**

    Girish*, Mohd Mujtaba Akhtar*, Orchid Chetia Phukan*, Drishti Singh, Swarup Ranjan Behera, Pailla Balakrishna Reddy, Arun Balaji Buduru, Rajesh Sharma

    EUSIPCO 2025

13. **ARE MAMBA-BASED AUDIO FOUNDATION MODELS THE BEST FIT FOR NON-VERBAL EMOTION RECOGNITION?**

    Mohd Mujtaba Akhtar*, Orchid Chetia Phukan*, ***Girish****, Swarup Ranjan Behera, Sanjib, Arun Balaji Buduru, Rajesh Sharma

    EUSIPCO 2025

14. **Are Multimodal Foundation Models All That Is Needed for Emofake Detection?**

    Mohd Mujtaba Akhtar*, ***Girish****, Orchid Chetia Phukan*, Swarup Ranjan Behera, Pailla Balakrishna Reddy, Ananda Chandra Nayak, Sanjib Kumar Nayak, Arun Balaji Buduru

    APSIPA 2025

15. **Rethinking Cross-Corpus Speech Emotion Recognition Benchmarking: Are Paralinguistic Pre-Trained Representations Sufficient?**

    Mohd Mujtaba Akhtar*, Orchid Chetia Phukan*, ***Girish****, Swarup Ranjan Behera, Sanjib, Arun Balaji Buduru, Rajesh Sharma

    APSIPA 2025

16. **Beyond Speech and More: Investigating the Emergent Ability of Speech Pre-Trained Models for Classifying Physiological Time-Series Signals**

    Orchid Chetia Phukan*, Swarup Ranjan Behera*, ***Girish****, Mohd Mujtaba Akhtar, Sanjib, Arun Balaji Buduru, Rajesh Sharma

    APSIPA 2025

17. **Investigating Polyglot Speech Foundation Models for Learning Collective Emotion from Crowds**

    Orchid Chetia Phukan*, ***Girish****, Mohd Mujtaba Akhtar*, Panchal Nayak, Priyabrata Mallick, Swarup Ranjan Behera, Parabattina Bhagath, Pailla Balakrishna Reddy, Arun Balaji Buduru

    APSIPA 2025

18. **Strong Alone, Stronger Together: Synergizing Modality-Binding Foundation Models with Optimal Transport for Non- Verbal Emotion Recognition**

    Orchid Chetia Phukan, Mohd Mujtaba Akhtar*, ***Girish****, Swarup Ranjan Behera, Sishir Kalita, Arun Balaji Buduru, Rajesh Sharma, SR Mahadeva Prasanna

    ICASSP 2025

19. **NeuRO: An Application for Code-Switched Autism Detection in Children**

    Mohd Mujtaba Akhtar*, ***Girish****, Orchid Chetia Phukan*, Muskaan Singh*

    INTERSPEECH Show & Tell 2024

Projects

Helix: Versatile AI Assistant GitHub

- AI-Powered Smart Responses: Uses OpenAI GPT, LangChain, and Pinecone for real-time, context-aware interactions, improving customer engagement and automation.
- Versatile & Scalable: Easily deploys across industries like healthcare, retail, and finance with minimal code changes, ensuring seamless adaptability.
- Omnichannel & Fast: Connects via WhatsApp (Twilio) and phone calls, with a FastAPI backend and async MongoDB for quick, efficient responses.
- **Tools Used: LangChain, FastAPI, MongoDB, Twilio.**

TwinVerify: Secure Encryption with Two-Factor Audio and Text Authentication Framework GitHub

- Created a unique audio encryption system to securely store answers in files, allowing access only for verified users.
- Developed a two-step authentication process using voice verification and text matching for safe encryption and decryption.
- Built a secure platform with random challenge questions, encrypted data storage, and easy file management.
- **Tools Used: Speech-to-Text, MongoDB, Natural Language Processing (NLP), Flask, C++**

Multimodal Personality Prediction Using Contrastive Learning GitHub

- Built a neural network with two processing paths that uses contrastive learning to predict personality traits (Openness, Conscientiousness, Extraversion, Agreeableness, Neuroticism) from audio and video data.
- Trained the model on both speech and visual cues, improving how it understands and learns personality traits from different sources.
- Optimized the system for better integration of multimodal data, ensuring accurate and reliable personality predictions from real-world audio and video inputs.
- **Tools Used: TensorFlow, Hugging Face Transformers.**

Golf Phase Detection and Analysis Application. GitHub

- Developed algorithms to detect and classify golf swing phases (setup, backswing, downswing, ball impact, and follow-through) with high accuracy.
- Designed a user-friendly interface for players and coaches, enabling easy review and real-time analysis across different video formats.
- **Tools Used: OpenCV, MediaPipe, FFmpeg, Streamlit**

Achievements

Certificates and Achievements:

- Awarded *First Runner-Up* in Technical Idea Presentation at ICE Day, UPES, Dehradun (22nd October 2024) for a concept related to speaker verification.
- Awarded a 1000 Euro ISCA travel grant to attend Interspeech 2025 in Rotterdam, The Netherlands.
- **Awarded the Social Impact Award at EACL 2026 for the paper** *"DIVINE: Coordinating Multimodal Disentangled Representations for Oro-Facial Neurological Disorder Assessment."*

Technical Skills

Frameworks and Libraries: TensorFlow, Keras, PyTorch, OpenCV, LangChain

Programming Languages: Python, C, Java, SQL

Web Technologies: HTML, JavaScript, FastAPI, Flask

Methodologies: Retrieval-Augmented Generation (RAG)

Technologies: Git, Linux, AWS, Docker

Volunteering

ISCA Team Volunteer:

- Managing script development and co-host episodes on speech technology and AI advancements.
- Reviewer for ICASSP 2026.

Academic & Industry Collaboration

- **Dr. Vidyanand Mishra:** Assistant Professor, UPES Dehradun, India
  E-mail: Vidyanand.mishra@ddn.upes.ac.in
- **Dr. Arun Balaji Buduru:** Associate Professor, IIIT-Delhi, India
  E-mail: arunb@iiitd.ac.in
- **Dr. Muskaan Singh:** Assistant Professor, Ulster University, UK
  E-mail: m.singh@ulster.uk.in
- **Dr. Rajesh Sharma:** Professor, Plaksha University, India
  E-mail: rajesh.sharma@plaksha.edu.in
- **Dr. Swarup Ranjan Bahera:** Data Scientist, ExxonMobil, India
  E-mail: swarupranjanbehera@gmail.com